

FoldShield⁺⁺ Structural Analysis Report

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Input Structures

Reference PDB	1UBQ.pdb
Query PDB	1CRN.pdb
Mode	ca
Ref chain	A
Query chain	A

Classical Coordinate Metrics

These metrics saturate near 1.0 even when function diverges. Use FoldShield++ signals below for structural perturbation insight.

Metric	Value	Note
TM-score	0.300	>=0.50 same fold · >=0.90 nearly identical
RMSD (A)	3.300	Lower is better · >3.0 A = significant deviation

FoldShield++ Signal Scores — Phase 1 + Phase 2

All active signals with scores, fusion weights, and interpretation. Phase 2 signals are highlighted in blue.

Braid similarity	0.673	0.20	moderate braid similarity
Motif similarity (UL-DSL)	0.586	0.37	moderate motif similarity
SSEF entropy (Phase 2) [P2]	0.714	0.20	moderate grammar rewrite — conformational transition regime
Persistent homology (Phase 3) [P2]	0.432	0.23	topological persistence features diverge

Fusion Layer v2 — Combined Score

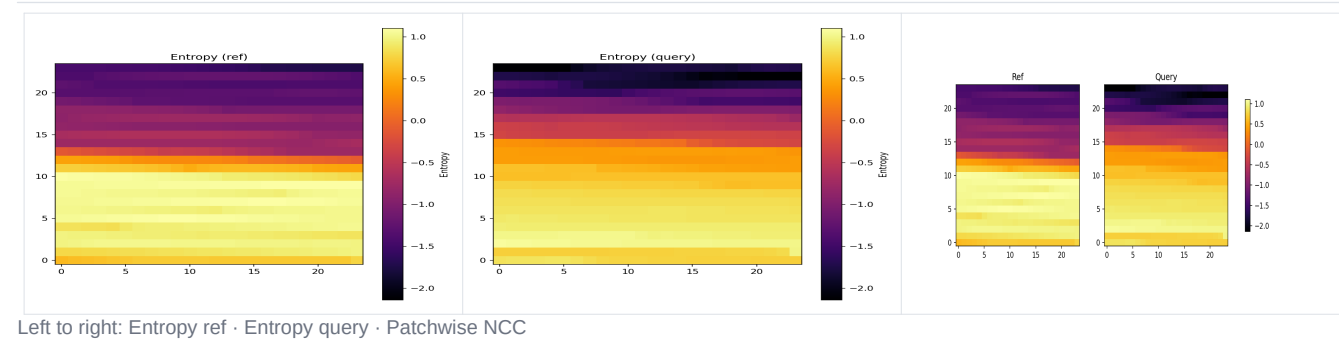
0.5940	phase2	moderately similar protein structures

Phase 2 — Perturbation Classification

SSEF score	Local topology	Gap score	Perturbation class
0.714	0.094	0.619	Conformational transition

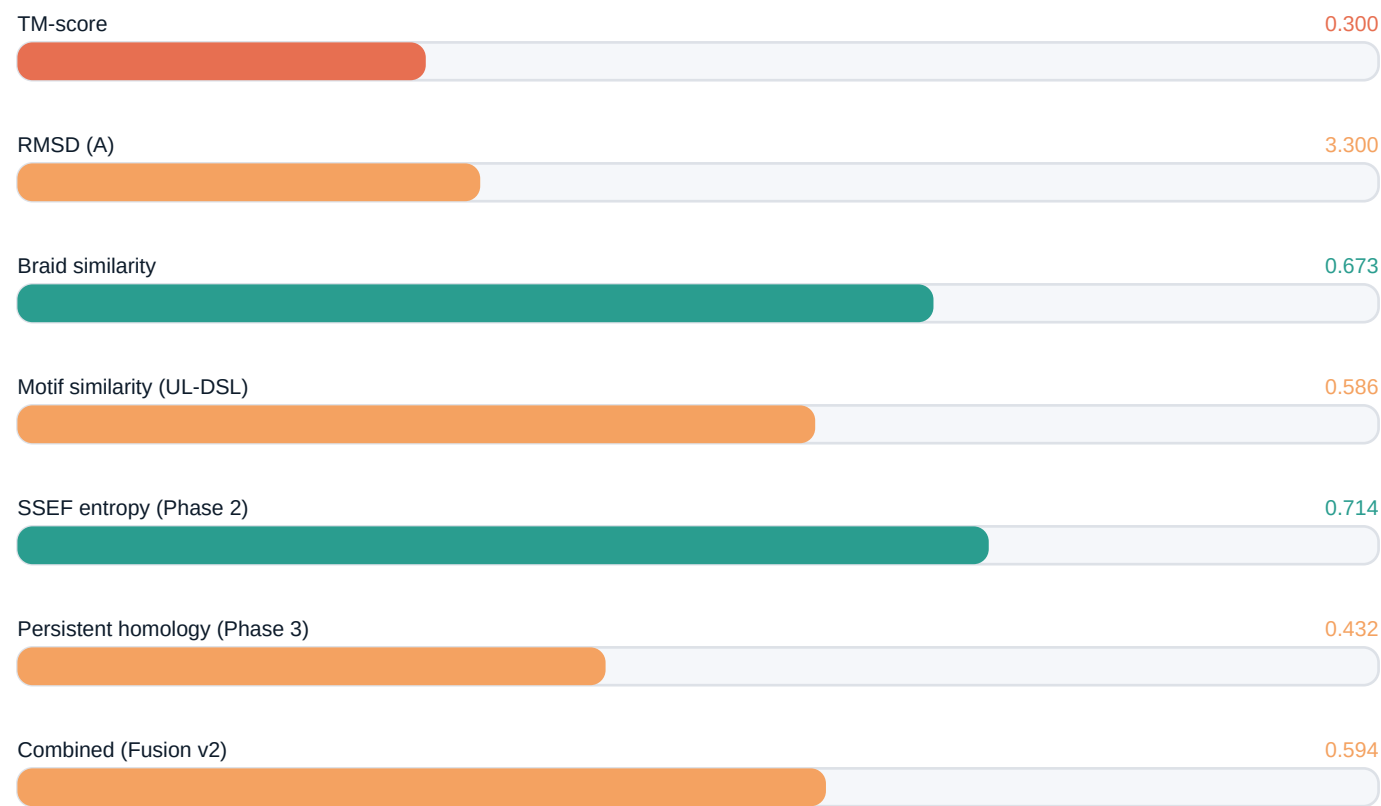
Gap = SSEF minus local topology. <0.15 buried-core mutation | 0.15-0.36 fold-switch | >0.36 conformational transition

Entropy Maps & Signal Profiles



Signal Score Bars

Visual 0-1 scale for all active signals. Green ≥ 0.66 / Amber 0.33-0.66 / Red < 0.33 .



Symbolic Score Detail

Symbolic Similarity Score	0.673	overall strong similarity
Behavioural Correlation Similarity	0.850	entropy maps strongly correlated
Structural Topology Similarity	0.600	partial generator overlap
Structural Topology Ordering Similarity	0.408	weak pattern similarity

FoldShield++ Phase 2 | RexCrux Research Laboratory | Signals: Braid / UL-DSL / Global Topology / Local Topology / SSEF Entropy | Fusion: Ridge Regression weighted linear combination

Signal Analysis — Figures

FoldShield++ | Signal Analysis | 1UBQ vs 1CRN

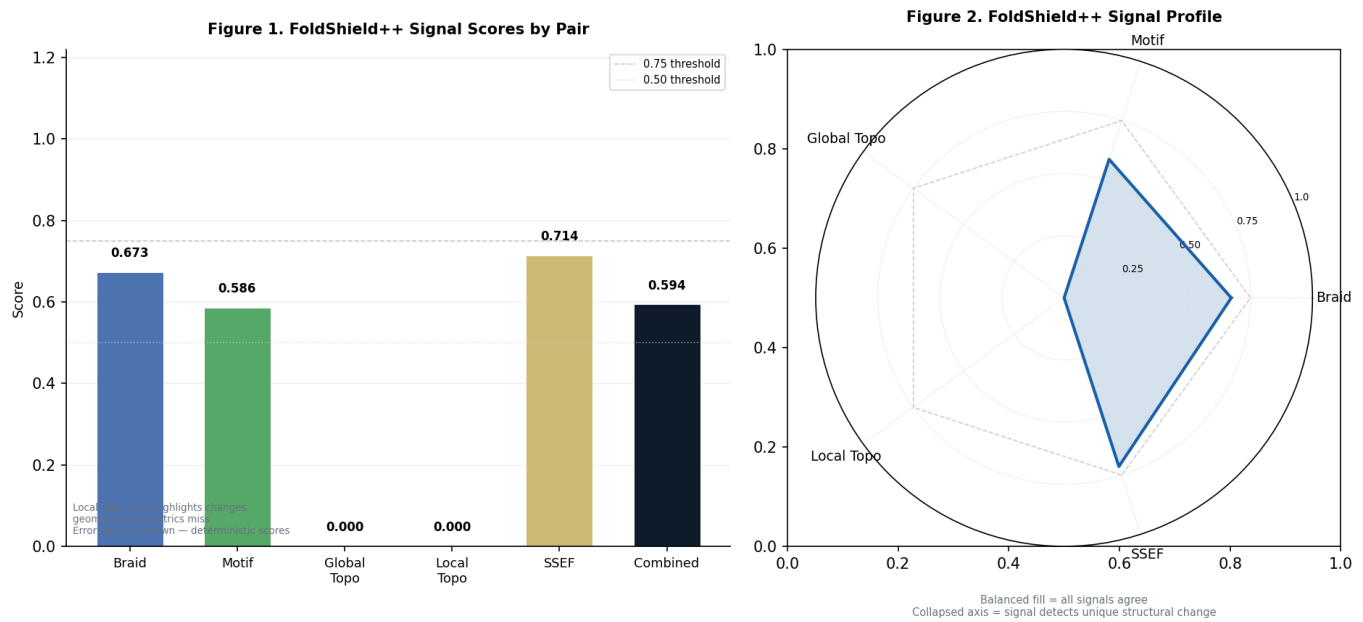


Figure 1: Bar chart comparing all five signals and the Combined score. Local Topo (red) highlights structural changes geometry-only metrics miss. Dashed = 0.75 · Dotted = 0.50. Error bars not shown — scores are deterministic. Figure 2: Radar chart showing the five-signal profile. Balanced fill = all signals agree. Collapsed axis = signal detects unique structural change.